

RADDAC: Risk-Adaptive Distributed Dynamic Access Control Framework for IoT Data Sharing

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Abstract—While securing data sharing is critical in boundary-blurred IoT environments, existing solutions remain insufficient against the potential risks of data interactions. To address these challenges, this paper proposes RADDAC, a risk-adaptive distributed dynamic access control framework for the IoT. First, it constructs a three-tier collaborative architecture comprising “Blockchain-Edge Computing-IPFS” to enhance the security and scalability of access control services. Furthermore, to tackle covert insider threats, RADDAC extends the standard Attribute-Based Access Control (ABAC) model by introducing a Risk Control Point (RCP). This component employs unsupervised clustering to establish a risk baseline and subsequently utilizes Kullback-Leibler (KL) divergence combined with joint attribute distributions to quantify access behavior risks, thereby establishing a closed-loop and continuous authorization mechanism. Comprehensive experiments demonstrate that RADDAC exhibits superior performance in terms of both security and efficiency.

Index Terms—Access control, Internet of Things (IoT), blockchain, data sharing, risk assessment.

I. INTRODUCTION

As a critical component of new digital infrastructure, the Internet of Things (IoT) is characterized by data mobility, which deeply empowers industrial applications. However, centralized IoT architectures inherently suffer from single points of failure and scalability bottlenecks. Furthermore, IoT environments typically house vast amounts of sensitive data; frequent cross-domain interactions further blur system security boundaries, significantly amplifying the risks of data tampering and leakage [1]. Consequently, ensuring trustworthy and controlled access to IoT data has emerged as a critical research focus.

Access control serves as a core mechanism for data security, ensuring that only authorized legitimate users can access data resources [2]. Traditional access control models, such as Discretionary Access Control (DAC) [3], Role-Based Access Control (RBAC) [4], and Attribute-Based Access Control (ABAC) [5], suffer from the rigidity of single static policies, making it difficult to meet the urgent demands for trustworthy and dynamic authorization in IoT environments. Moreover, these schemes [3]–[5] continue to rely on centralized access control architectures, leaving them vulnerable to single points of failure.

Blockchain technology offers significant security advantages for IoT data sharing [6]. While blockchain-integrated access control schemes excel at defending against external attacks, they lack effective mechanisms to thwart data

theft perpetrated through means such as social engineering. Statistics indicate that insider threats have become a primary factor in data breaches [7]. Unfortunately, existing blockchain-based IoT access control schemes [8]–[10] typically employ static, one-time policy verification and lack in-depth analysis of access behavior risks, rendering them inadequate against internal threats.

To address these challenges, this paper proposes RADDAC, a distributed risk-adaptive dynamic access control scheme for the IoT. The main contributions are summarized as follows:

- 1) We construct a “Blockchain-Edge Computing-IPFS” three-tier collaborative architecture. The integration of blockchain enhances the security and scalability of the access control architecture, while smart contracts enable automated and efficient access control decisions and auditing.
- 2) We extend the standard ABAC model by introducing a Risk Control Point (RCP). By incorporating risk factors into the decision-making process, we establish a dynamic closed-loop access control mechanism spanning from dynamic decision-making to adaptive adjustment, thereby achieving dynamic, continuous, and fine-grained authorization.
- 3) We propose a data-driven adaptive risk quantification mechanism. This mechanism employs unsupervised clustering to determine risk baselines, followed by the use of Kullback-Leibler (KL) divergence combined with joint attribute distributions to quantify access behavior risks.
- 4) Comprehensive experimental evaluations validate the superiority of the proposed scheme in terms of both performance and effectiveness.

II. RELATED WORK

Access control serves as the core mechanism for safeguarding data security and preventing unauthorized access. However, traditional access control models each possess inherent limitations. Discretionary Access Control (DAC) allows resource owners to manage access autonomously, offering simplicity and flexibility but lacking fine-grained control. Mandatory Access Control (MAC) enforces strict multi-level security policies but suffers from insufficient flexibility. Role-Based Access Control (RBAC) faces limited scalability in

dynamic IoT environments due to its reliance on pre-defined roles. While Attribute-Based Access Control (ABAC) achieves fine-grained authorization, its typically centralized decision-making architecture poses a risk of single points of failure. Although scheme [11] constructs a distributed access control architecture based on ABAC, it lacks dynamic capabilities and security assessments.

Blockchain technology provides a trusted foundation for secure data interaction among IoT devices [12], leading researchers to increasingly explore the deep integration of blockchain and access control. Han et al. [9] categorized data into public data (stored in the cloud) and private data (stored on the blockchain), utilizing on-chain access policies and auditing mechanisms to ensure private data security; however, this imposes heavy computational and storage burdens on the blockchain. To address this issue, Liu et al. [13] adopted a method combining on-chain permission control with off-chain data storage to alleviate the storage load. To optimize authorization capabilities, He et al. [14] integrated RBAC and ABAC models, enhancing permission management flexibility through a role-attribute hybrid mechanism. Nevertheless, similar to Kamboj et al. [15], they face the challenge of uniformly defining roles within heterogeneous IoT environments. Saini et al. [16] designed an on-chain contract revocation mechanism to promptly block unauthorized access, yet they did not elucidate the mechanism for determining illegal behavior.

To overcome the limitations of traditional access control relying solely on pre-set permissions, Saha et al. [17] incorporated risk factors into authorization decisions within blockchain access control. They quantified access requests across multiple dimensions, including operational, security, and heuristic risks, but did not detail the specific risk quantification methods. Wang et al. [18] identified risky access by modeling the temporal relationships and task logic of control processes, but it remains unclear how to assess and adjust the temporal model at runtime. Akhuseyinoglu et al. [19] proposed a dynamic access control scheme for cyber-physical systems based on ABAC, employing a Goal-Action-Generation Model (GAGM) for risk assessment. However, their risk assessment model relies on pre-defined, scenario-based expert knowledge or experience, leading to significant subjectivity. Furthermore, the reputation mechanism adopted in scheme [20] derives scores entirely from user mutual evaluations, making it dependent on subjective feedback and susceptible to malicious collusion.

III. PROPOSED SCHEME

This section introduces the system framework of the proposed scheme and the improved access control model.

A. System Framework

The overall framework design of RADDAC is illustrated in Fig. 1, comprising six core entities: 1) IoT Devices: Responsible for environmental sensing and data collection. 2) Edge Servers: Acting as device managers and blockchain nodes, they are responsible for data preprocessing and uploading

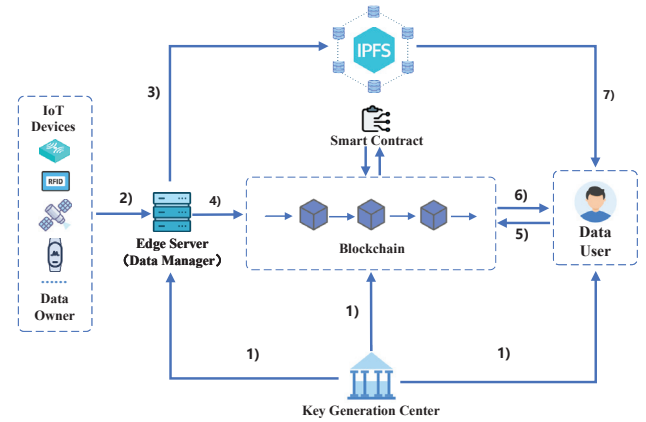


Fig. 1: System framework

to the blockchain, while supporting dynamic scalability. 3) Consortium Blockchain: Serving as the access control execution platform, it is tasked with storing metadata and making access authorization decisions. 4) Data Users: Users who require access to IoT data. 5) IPFS: Provides low-cost, scalable distributed storage to offload the storage burden from the blockchain. 6) KGC: Acts as a trusted third party responsible for unified key management.

The data sharing workflow of this framework consists of seven steps: 1) The KGC initializes the system and distributes keys to each entity; 2) IoT devices collect data and transmit it to edge server nodes after initial processing; 3) Edge servers preprocess the data, upload it to IPFS, and obtain the data index (Content Identifier, CID) from IPFS; 4) Subsequently, metadata (containing data descriptions, CIDs, etc.) and access policies are uploaded to the blockchain via smart contracts; 5) Data users initiate access requests to the blockchain; 6) The access control contract automatically executes the authorization decision; upon authorization, it returns the corresponding metadata to the data requester; 7) The data requester retrieves the required data from IPFS using the CID within the metadata.

B. Risk-Adaptive Dynamic Access Control Model

As illustrated in Fig.2, RADDAC modifies and extends the standard ABAC architecture.

The standard ABAC model primarily consists of an Attribute Authority (AA), Policy Administration Point (PAP), Policy Decision Point (PDP), Policy Information Point (PIP), and Policy Enforcement Point (PEP). RADDAC utilizes the blockchain as the AA to ensure the trustworthiness of access control information. Furthermore, incorporating the concepts of attribute mutability and authorization continuity [21], RADDAC introduces a Risk Control Point (RCP). By combining four stages—baseline determination, risk pre-check, behavioral risk assessment, and active feedback—it constructs a dynamic closed-loop access control mechanism centered on “access decision - active feedback - adaptive adjustment”. The core advantage of RADDAC lies in utilizing mutable risk

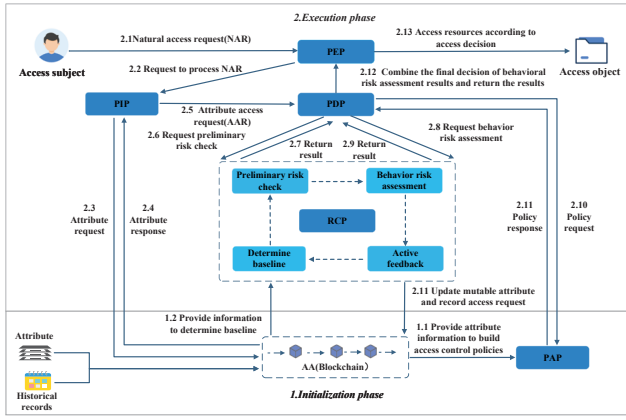


Fig. 2: RADDAC access control model

attributes to achieve dynamic risk perception and closed-loop feedback on access behaviors. This transforms access control from a static “request-response” process into a dynamic adaptive adjustment process, effectively overcoming the limitation of traditional ABAC in proactively addressing insider threats.

Based on the aforementioned extensions, the implementation of RADDAC is divided into two phases: initialization and execution.

1) **Initialization Phase:** This phase establishes objective standards for the dynamic decision-making of RADDAC and primarily consists of two steps: a) Policy Construction and b) Dynamic Baseline Determination.

a) **Policy Construction:** The PAP maintains a policy set based on the attribute information provided by the PIP. We define the access request attribute set as $A = \{sAttr, oAttr, cAttr\}$, where $sAttr$, $oAttr$, and $cAttr$ denote the attribute characteristics of the access subject, access object, and dynamic context (e.g., time, frequency), respectively.

b) **Risk Baseline Determination:** In actual IoT environments, significant heterogeneity exists in the business logic of different organizational units. For instance, in a smart factory, the access frequency and characteristics of the production monitoring department differ distinctly from those of the warehousing and logistics department. A single global baseline would not only fail to accurately capture the specific risk characteristics of each department but also impose a heavy computational burden on the blockchain. Therefore, this scheme adopts a department-based risk quantification strategy, where the department administrator periodically (e.g., daily or weekly) invokes the smart contract $RADDAC_SC$ to determine the risk baseline for that department.

To quantify the departmental baseline, we first perform standardized modeling on the historical access behaviors of all access subjects $\{s_1, s_2, \dots, s_k\}$ within the department. For any subject s_i , we extract their recent n access records and map them into an m -dimensional Boolean feature vector:

$$\mathbf{v}_{s_i}^{(t)} = [v_{s_i,1}^{(t)}, v_{s_i,2}^{(t)}, \dots, v_{s_i,m}^{(t)}], \quad v_{s_i,k}^{(t)} \in \{0, 1\} \quad (1)$$

where $v_{s_i,k}^{(t)}$ represents the status of the t -th access on the k -th dimension attribute. Furthermore, to obtain the stable

behavior characteristics of the subject within the current cycle, we perform temporal logical aggregation on the record set. We employ a bitwise logical OR operation to extract the union of features, generating the feature vector $\mathbf{V}_{s_i}$ for that subject, defined as follows:

$$\mathbf{V}_{s_i} = [V_{s_i,1}, V_{s_i,2}, \dots, V_{s_i,m}], \quad V_{s_i,k} = \bigvee_{t=1}^n v_{s_i,k}^{(t)} \quad (2)$$

After obtaining the aggregated feature vector set $\mathcal{V} = \{\mathbf{V}_{s_1}, \mathbf{V}_{s_2}, \dots, \mathbf{V}_{s_k}\}$ for all subjects in the department, given the unknown nature of access behavior patterns, we adopt the unsupervised Density Peak Clustering (DPC) algorithm [22] to partition the subjects. This algorithm utilizes the distance metric between $\mathcal{V}$ to divide access subjects into a normal subject set C_N and an anomalous subject set C_{AN} .

Based on the statistical distribution characteristics of C_N , we derive the subject risk threshold $Risk_B^S(S)$ and the access request behavior risk threshold $Risk_B^R(S)$ according to the mean μ_N and standard deviation σ_N of the risk attribute values of subjects in C_N , along with a confidence coefficient k :

$$\begin{cases} Risk_B^S(S) = \mu_N^S + k \cdot \sigma_N^S \\ Risk_B^R(S) = \mu_N^R + k \cdot \sigma_N^R \end{cases} \quad (3)$$

In addition, we need to determine the behavioral baseline attribute distribution based on the historical access request attribute distribution of subjects in C_N , to serve as a basis for subsequently quantifying the risk value of access requests. Given the sparsity of the high-dimensional Boolean feature space, we construct three key joint attribute distributions to approximately characterize the mainstream access patterns of the department: Subject-Object distribution $Q(sAttr, oAttr)$, Subject-Context distribution $Q(sAttr, cAttr)$, and Object-Context distribution $Q(oAttr, cAttr)$.

Taking the Subject-Object joint distribution as an example, the calculation is as follows:

$$Q(sAttr, oAttr) = \frac{\sum_{AR \in C_N} (sAttr, oAttr) + \epsilon}{Total(AR \in C_N) + \epsilon \cdot |\Omega|} \quad (4)$$

where $\sum_{AR \in C_N} (sAttr, oAttr)$ counts the occurrence frequency of specific subject-object attribute combinations in C_N , $Total(AR \in C_N)$ denotes the total number of all access requests AR in C_N , $|\Omega|$ is the total number of all possible attribute combinations, and ϵ is the smoothing factor.

2) **Execution Phase:** The access subject s_i initiates a data access request by invoking the smart contract $RADDAC_SC$. The system executes a multi-stage comprehensive authorization decision based on the preset departmental risk thresholds and the behavioral baseline attribute distributions, as shown in Algorithm 1.

a) **Access Subject Risk Pre-check:** The system first encapsulates the access request into an Attribute-Based Access Request (AAR) based on the request’s dynamic context. To rapidly intercept high-risk users, the system first verifies the access subject’s current risk attribute value $Risk^S(s_i)$. If this value exceeds the subject risk threshold $Risk_B^S(S)$ determined in the initialization phase, access is immediately

Algorithm 1 Execution Phase

Input: AR **Output:** Permit / Deny

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1:  $AAR = PreprocessRequest(AR)$ 
2: if  $Risk^S(s_i) \geq Risk_B^S(S)$  then
3:   return Deny
4: end if
5:  $P \leftarrow GetBehaviorDistribution(ARs \in s_i)$ 
6: for  $(x, y) \in \{sAttr, oAttr, cAttr\}$ ,  $x \neq y$  do
7:    $K(x, y) \leftarrow CalcKL(P(x, y), Q(x, y))$ 
8: end for
9:  $Risk^R(s_i) = K(s, o) + K(s, c) + K(o, c)$ 
10: if  $Match(AAR, policy) = \text{true}$  then
11:    $updateAndSave(s_i, AR)$  & return Permit
12: else
13:    $updateAndSave(s_i, AR)$  & return Deny
14: end if

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denied; otherwise, the process proceeds to the dynamic risk assessment stage.

$$result_{pre} = \begin{cases} \text{next step,} & \text{if } Risk^S(s_i) < Risk_B^S(S) \\ \text{deny,} & \text{otherwise} \end{cases} \quad (5)$$

b) Dynamic Access Behavior Risk Assessment: Upon passing the pre-check, the system quantifies the degree to which the current access behavior deviates from the departmental access behavior baseline. First, based on the recent access records of subject s_i , the system constructs the measured attribute distribution P that reflects their recent access behavior characteristics. Taking the subject-object measured attribute distribution $P(sAttr, oAttr)$ as an example, the calculation is as follows:

$$P(sAttr, oAttr) = \frac{\sum_{AR \in s_i} (sAttr, oAttr) + \epsilon}{Total(AR \in s_i) + \epsilon \cdot |\Omega|} \quad (6)$$

where $\sum_{AR \in s_i} (sAttr, oAttr)$ counts the frequency of specific attribute combinations for subject s_i in the recent period, and $Total(AR \in s_i)$ denotes the total number of their recent requests.

Subsequently, Kullback-Leibler (KL) divergence is employed to measure the discrepancy between the departmental access behavior attribute baseline distribution Q and the access behavior attribute measured distribution P of s_i . A larger discrepancy implies a higher “unexpected information content” contained in the access behavior of s_i , indicating a greater access risk. Taking the subject-object joint distribution as an example, its KL divergence calculation is:

$$K(sAttr, oAttr) = \sum P(sAttr, oAttr) \times \log \frac{P(sAttr, oAttr)}{Q(sAttr, oAttr)} \quad (7)$$

Finally, the current access behavior risk value $Risk^R(s_i)$ of s_i is derived from the sum of the KL divergences of the three

key joint attribute distributions:

$$Risk^R(s_i) = K(sAttr, oAttr) + K(sAttr, cAttr) + K(oAttr, cAttr) \quad (8)$$

c) Comprehensive Authorization Decision: The system performs a comprehensive authorization in conjunction with the dynamic risk assessment results. If the access policy matches successfully, access authorization is granted; otherwise, authorization is denied. The system subsequently executes the authorization based on the decision result and responds to the access subject s_i with metadata information.

$$result_2 = \begin{cases} \text{permit,} & \text{if } Match(AAR, policy) \\ \text{deny,} & \text{otherwise} \end{cases} \quad (9)$$

d) Active Feedback: Following the comprehensive decision, the system requires the dynamic update of the access subject's risk attribute value $Risk^S(s_i)$ and the immutable recording of access log information on the blockchain.

To smoothly reflect the changing trends in the subject's recent behavior, we employ the Exponential Weighted Moving Average (EWMA) method to update the subject's risk value:

$$Risk^S(s_i)_n = \lambda \cdot Risk^R(s_i) + (1 - \lambda) \cdot Risk^S(s_i)_{n-1} \quad (10)$$

$$\lambda = \min \left(\lambda_{base} + \alpha \cdot \frac{\Delta R}{Risk^S(s_i)_{n-1}}, 1 \right) \quad (11)$$

where λ_{base} is the base smoothing factor, α is the sensitivity coefficient, and ΔR represents the difference between the current access behavior risk value $Risk^R(s_i)$ and the previous access request behavior risk value $Risk^S(s_i)_{n-1}$.

Furthermore, information related to the current request is recorded on the blockchain. These trusted access logs serve not only for post-event auditing but also as historical data of individual behavior to dynamically adjust access patterns. Consequently, each access instance becomes a historical sample that influences future authorizations.

In summary, RADDAC constructs a closed-loop access control mechanism spanning “dynamic decision-making, active feedback, and adaptive adjustment,” achieving dynamic adaptability for the entire access control system.

IV. EXPERIMENTAL ANALYSIS

A. Implementation Environment

To evaluate the performance and effectiveness of RADDAC, we implemented a distributed access control prototype for data sharing based on the Hyperledger Fabric platform. The experiments were conducted on a PC running Ubuntu 20.04 LTS, equipped with an Intel Core i7-13650HX CPU @2.60 GHz and 32 GB of RAM. The experimental network adopts a consortium blockchain architecture to simulate data sharing scenarios. All network nodes are containerized and orchestrated using Docker Compose to ensure low-latency inter-node communication. The core access control logic of RADDAC is encapsulated within a smart contract, denoted as *RADDAC_SC*, which is developed in Golang. This contract is deployed and instantiated to execute trusted on-chain authorization decisions.

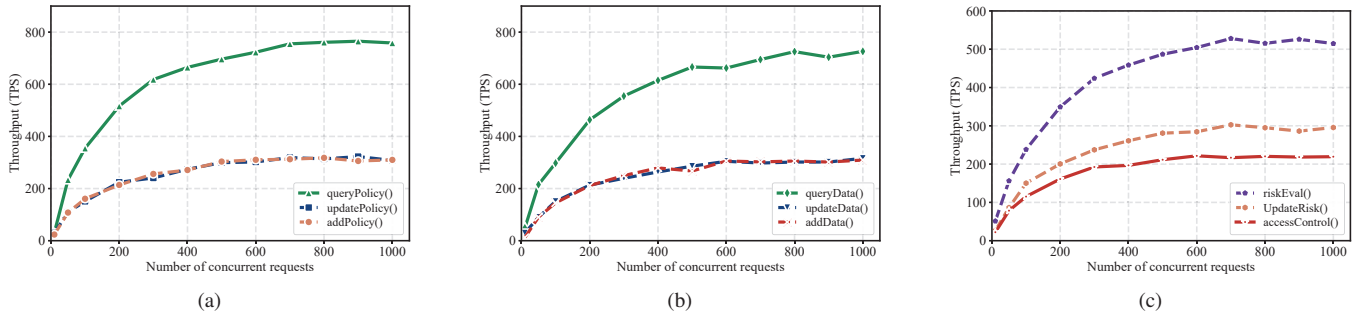


Fig. 3: Throughput performance of different system functionalities under varying numbers of concurrent requests.

B. Results And Discussion

1) *Throughput*: First, the experiment evaluated the system's throughput performance. Hyperledger Caliper was utilized to generate concurrent requests, testing the system's throughput under varying pressures by dynamically adjusting the transaction send rate. As shown in Fig. 3, overall, as the client request send rate increases, the throughput of each tested function sequentially reaches saturation. In terms of specific functions, *addData()* and *updatePolicy()* involve write operations that necessitate consensus across network nodes, resulting in lower throughput compared to read operations. Meanwhile, *accessControl()*, as the function responsible for receiving and processing access requests, involves extensive computational processing, thus exhibiting the lowest throughput. Furthermore, throughput is primarily constrained by the host machine and computational resources; increasing the number of Peer nodes or upgrading resource configurations could further enhance throughput.

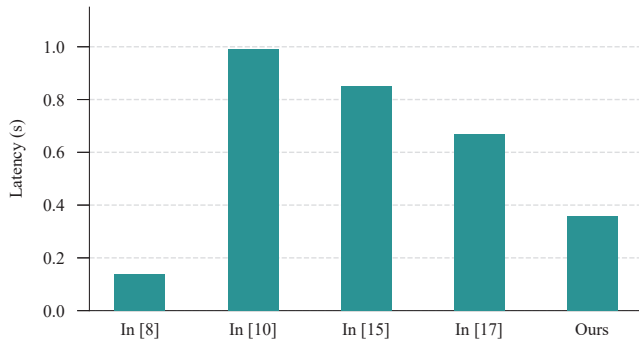


Fig. 4: Latency comparison with other schemes

2) *Latency*: Regarding the transaction processing latency metric (the time interval from request submission to transaction confirmation), we employed Caliper to generate access requests and utilized a constant-rate load controller to maintain network stability, thereby testing the system's execution latency. As shown in Fig. 4, RADDAC exhibits low latency. However, compared to scheme [8], RADDAC's access control decision latency is slightly increased. The reason is that [8] simply employs static access control policies for authorization decisions, whereas RADDAC executes a multi-stage comprehensive decision process involving risk pre-checks and

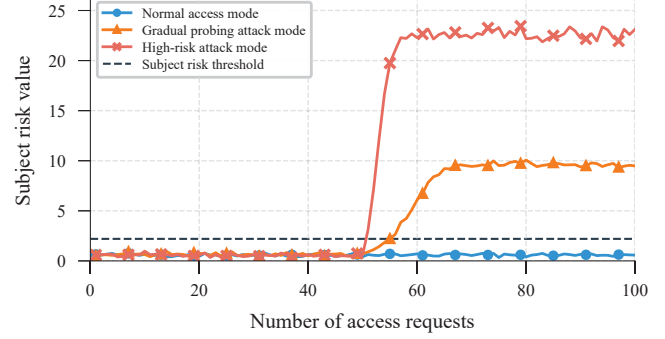


Fig. 5: Risk variations across three access models

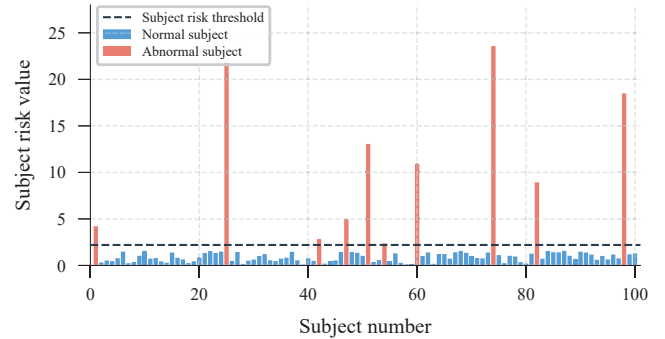


Fig. 6: Subject risk values in an example department

behavioral risk assessments during authorization. This acceptable performance overhead is exchanged for a significant enhancement in decision security.

3) *Effectiveness Evaluation*: To verify the effectiveness of RADDAC in detecting progressive probing attacks, we designed three access modes for validation: a) Normal Access Mode: The access subject strictly adheres to business logic and accesses data within their permissions; b) Progressive Probing Attack Mode: The attacker employs low-amplitude unauthorized access strategies to gradually probe system boundaries; c) High-risk Malicious Attack Mode: The attacker implements overt unauthorized data access. In the experiment, the first 50 requests were set to the normal access mode, with attacks injected starting from the 51st request. As shown in Fig. 5, in mode 1), the subject's risk value consistently remains below

the threshold. In mode 2), the subject's risk value does not rise significantly initially; however, as access requests increase, the risk value grows rapidly and exceeds the threshold. In mode 3), due to the subject's access behavior pattern deviating significantly from the baseline, the risk value exceeds the threshold within a short period and sustains a high level.

As illustrated in Fig. 6, after a period of access, taking the Quality Inspection Department of a smart factory as an example, the risk values of all anomalous access subjects exceed the access subject risk threshold. It is worth noting that, to more intuitively demonstrate the effect of RADDAC's dynamic risk assessment, we disabled the subject risk verification in the risk pre-check phase for this specific visualization. In actual deployment, when an access subject's risk value exceeds the threshold for the first time, the system marks them as "pending review" and rejects subsequent requests. This mechanism enables RADDAC to achieve early and effective blocking of malicious access.

V. CONCLUSION

This paper proposes RADDAC, a risk-adaptive access control scheme for IoT data sharing. By constructing a "Blockchain-Edge Computing-IPFS" collaborative architecture, the scheme ensures system scalability and decision trustworthiness. Furthermore, by introducing a Risk Control Point (RCP), it establishes a closed-loop mechanism characterized by "dynamic decision-making, active feedback, and adaptive adjustment" to mitigate insider threats. Specifically, this paper utilizes unsupervised clustering and multi-attribute joint distributions to achieve objective risk quantification, thereby enhancing the accuracy of risk assessment. Comprehensive experiments demonstrate that RADDAC significantly improves security while guaranteeing efficient decision-making. Future work will focus on enhancing the scheme's performance by leveraging off-chain computation mechanisms.

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